

A piecewise-homogeneous tracial case of the positive-projection conjecture

Autonomous proof packet for arXiv:1909.00391

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Problem and result

Conjecture 6.1 of Arhancet asks, for $1 < p < \infty$, $p \neq 2$, whether the positively contractively complemented subspaces of Haagerup noncommutative L^p -spaces are exactly the nonassociative L^p -spaces of JW^* -algebras with faithful normal states. The exact source statement is reproduced below.

6 Open questions

It is natural to state the following conjecture in view of our results. In a next version of this preprint, we will hope finish the case where $\mathcal{M}$ is a finite von Neumann algebra.

Conjecture 6.1 *Suppose $1 < p < \infty$ with $p \neq 2$. Let X be a Banach space. Then X is isometric to a positively contractively complemented subspace of a Haagerup noncommutative L^p -space $L^{p,H}(\mathcal{M}, \varphi)$ where $\mathcal{M}$ is a σ -finite von Neumann algebra equipped with a normal faithful state φ if and only if X is isometric to a nonassociative L^p -space associated with a JW^* -algebra equipped with a normal faithful state.*

The later companion paper [2] defines $L^p(\mathcal{F}, \tau) = (\mathcal{F}, \mathcal{F}_*)_{1/p}$ using the trace-induced compatibility and proves the embedding implication for a single tracial JW^* -factor with separable predual. We extend that implication across a substantial class of nonfactorial algebras.

Theorem 1 (piecewise-homogeneous tracial embedding). *Let $1 \leq p < \infty$. For each j in a finite or countable set I , let (Ω_j, μ_j) be a finite measure space and let $\mathcal{F}_j$ be a JW^* -factor with separable predual and normalized normal faithful trace τ_j . Put*

$$\mathcal{J} = \bigoplus_{j \in I}^{\ell^\infty} L^\infty(\Omega_j, \mu_j; \mathcal{F}_j)$$

and equip it with the normal finite faithful trace

$$\tau(x) = \sum_{j \in I} a_j \int_{\Omega_j} \tau_j(x_j(\omega)) d\mu_j(\omega), \quad a_j > 0,$$

where the weights make $\tau(1) < \infty$. Then the nonassociative space $L^p(\mathcal{J}, \tau)$ is isometric to the range of a positive contractive projection on $L^p(\mathcal{M}, \sigma)$ for a finite von Neumann algebra $(\mathcal{M}, \sigma)$.

Thus the embedding direction of Conjecture 6.1 holds for all such tracial $\mathcal{J}$. This includes countable atomic centers (Ω_j singletons) and homogeneous diffuse centers.

Proof

Fix j . By Theorem 4.7 of [2], there are a finite von Neumann algebra $(\mathcal{N}_j, s_j)$ and a positive contractive projection

$$R_j : L^p(\mathcal{N}_j, s_j) \longrightarrow L^p(\mathcal{N}_j, s_j)$$

whose range is isometric to $L^p(\mathcal{F}_j, \tau_j)$. In the construction of that theorem the projection is induced by a trace-preserving normal positive projection at the algebra level; equivalently, it may be applied pointwise.

Set

$$\mathcal{M}_j = L^\infty(\Omega_j, \mu_j) \bar{\otimes} \mathcal{N}_j, \quad \sigma_j = \mu_j \otimes s_j.$$

The induced map

$$\tilde{R}_j f(\omega) = R_j(f(\omega))$$

is a positive contractive projection on $L^p(\mathcal{M}_j, \sigma_j) = L^p(\Omega_j; L^p(\mathcal{N}_j, s_j))$. Its range is

$$L^p(\Omega_j; L^p(\mathcal{F}_j, \tau_j)). \quad (1)$$

On the other hand, exact vector-valued complex interpolation gives

$$\begin{aligned} L^p(L^\infty(\Omega_j; \mathcal{F}_j), \mu_j \otimes \tau_j) &= (L^\infty(\Omega_j; \mathcal{F}_j), L^1(\Omega_j; (\mathcal{F}_j)_*))_{1/p} \\ &= L^p(\Omega_j; (\mathcal{F}_j, (\mathcal{F}_j)_*)_{1/p}) \\ &= L^p(\Omega_j; L^p(\mathcal{F}_j, \tau_j)) \end{aligned} \quad (2)$$

isometrically. Hence (1) is exactly the desired homogeneous nonassociative L^p -space, not merely an isomorphic copy with uncontrolled constants.

Now take the finite-tracial von Neumann algebraic direct sum $\mathcal{M} = \bigoplus_{j \in I}^{\ell^\infty} \mathcal{M}_j$ and give its j th summand the central trace weight a_j . Standard central decomposition yields

$$L^p(\mathcal{M}, \sigma) = \left(\bigoplus_{j \in I} L^p(\mathcal{M}_j, \sigma_j) \right)_{\ell^p(a_j)}.$$

The coordinatewise map $R = \bigoplus_j \tilde{R}_j$ is positive and idempotent, and

$$\|Rx\|_p^p = \sum_j a_j \|\tilde{R}_j x_j\|_p^p \leq \sum_j a_j \|x_j\|_p^p = \|x\|_p^p.$$

Its range is the same weighted ℓ^p -sum of the spaces in (2). Applying the identical central-sum interpolation formula to $(\mathcal{J}, \mathcal{J}_*)$ identifies that range isometrically with $L^p(\mathcal{J}, \tau)$. This proves the theorem. $\square$

Proof intuition

There are two exact stability principles. A factor solution survives adding a scalar measure variable because complex interpolation commutes with $L^p(\Omega; \cdot)$. It also survives a countable central sum because both the ambient and nonassociative L^p norms become the same weighted ℓ^p norm. Positivity and contractivity can therefore be checked one fiber at a time.

The point is not just closure under finite direct sums: diffuse homogeneous centers are included as well. What remains outside the theorem is a genuinely varying measurable field of non-isomorphic factors, where one needs a measurable choice of ambient von Neumann algebra and factor projection.

Scope and remaining obstruction

This does not prove Conjecture 6.1. The companion paper itself leaves the arbitrary tracial algebra as Conjecture 5.3 and notes that the direct-integral formalism is unclear. Our argument removes this issue only for countably piecewise-constant fields. For nonracial states there is a further modular obstruction: a state-preserving Jordan conditional expectation need not be selfadjoint, and state-independence of $(\mathcal{J}, \mathcal{J}_*)_{1/p}$ is itself open. Finally, the general forward implication still requires identifying the range of the lifted positive projection with the appropriate interpolation space.

References

- [1] C. Arhancet, *Positive contractive projections on noncommutative L^p -spaces and nonassociative L^p -spaces*, arXiv:1909.00391v8.
- [2] C. Arhancet, *Nonassociative L^p -spaces and embeddings in noncommutative L^p -spaces*, arXiv:2307.04452v3, especially Theorem 4.7 and Conjecture 5.3.